제목

Controlled Glot500-Style Replay with Vocabulary-Extension Initialization

- 92 XLM-R-seen Glot500 language-scripts
- 10 Glot500-internal non-XLM-R target language-scripts
- novelty: appended vocabulary row initialization

Source:
docs/exp/v5/4_reporting/03_final_report/claim_ledger.md
docs/exp/v5/4_reporting/final_claim_decision_tree.md

문제의식과 기여

Problem: multilingual coverage is broad, but not uniform.

- Reproduction: artifact-audited 92+10 Glot500-style replay
- Novelty: appended vocabulary row initialization with FVT
- Fidelity: PPPL, retrieval, classification, tagging, Roundtrip retained
- Claim rule: final method claims wait for matched checkpoints

재연 범위

Full Glot500 reproduction 이 아니라 controlled subset replay 이다 .

Scope comparison:

- scale: Glot500 500+ languages -> v5 102 language-scripts
- corpus: Glot500-c -> 92 seen + 10 target
- tokenizer: SentencePiece expansion -> SPM append-style expansion
- training: continued MLM -> continued MLM
- evaluation: Glot500 metrics -> same metric families

Backup evidence:

docs/exp/v5/4_reporting/00_tables/table_15_glot500_reproduction_fidelity.md
docs/exp/v5/4_reporting/01_figures/generated/figure_01_experiment_pipeline.png

금지 표현 :

full 511-language Glot500 reproduction

Target10 선정

Target10 은 Glot500 내부 raw dataset 에서 골랐다 .
선정 조건 :
• XLM-R seen 이 아님
• new_length >= 30000
• local raw directory 존재
• 지역 / 문자 / 어족 다양성
Coverage snapshot:
• scripts: Latn, Cyrl, Arab, Tibt, Olck
• regions: Europe, North Caucasus, West Asia, Himalaya, South/Southeast Asia, West Africa, Mesoamerica, Andean South America, Polynesia
• new_length range: 30,052-52,732
Target codes:
fur_Latn, krc_Cyrl, acm_Arab, dzo_Tibt, sat_Olck,
mad_Latn, bam_Latn, kjb_Latn, quw_Latn, rap_Latn
Source:
docs/exp/v5/0_tokenizer/miscellaneous/glot50010_selected_manifest.tsv

Corpus construction

Data scope 는 고정됐다 .
Corpus snapshot:
- seen languages: 92
- target languages: 10
- merged lines: 92,452,251
- output size: 19G
- missing language dirs: 0

Claim:
The 102-language corpus is reproducibly merged with no missing directories.

Tokenizer method

v5 는 Glot500-style SentencePiece append route 를 따른다 .

- XLM-R SPM 에 auxiliary SPM 의 novel pieces 를 append
- extended HF tokens: 368,687
- appended token strings: 118,685
- <mask> id: 250001 -> 368686

Audit result:

- 29/30 audited languages improved
- target10: 9/10 improved
- dzo_Tibt: 4.223938 -> 5.552124 tokens/word 로 악화

Figure:

docs/exp/v5/4_reporting/01_figures/generated/figure_02_tokenizer_fertility_delta.png

Novelty

Novelty 는 corpus 가 아니라 appended vocabulary row initialization 이다 .
Initialization arms:

- random: HF default random resize, baseline
- mean: source/global mean, stable ablation
- fvt: source-token decomposition mean, main novelty
- align: script-aware fallback, exploratory

핵심 질문 :
new token row 를 random 으로 둘 것인가 , 기존 tokenizer decomposition 으로 초기화할 것인가 ?

Initialization audit

초기화 비교는 row-copy correctness 가 보장될 때만 의미가 있다 .
필수 audit:
• source rows copied by token identity
• <mask> row remapped explicitly
• byte fallback rows separated
• LM head tying verified
Main FVT audit:
• source identity rows: 250,002
• FVT rows: 118,427
• byte/global-mean rows: 256
• lexical fallback rows: 2
• <mask> max abs diff: 0.0

Zero-step evidence

FVT 는 training 전 target MLM proxy 에서 가장 좋다 .
Zero-step target weighted NLL:
- random: 18.411756
- mean: 11.953142
- fvt: 8.785518

FVT vs random:
- target: -9.626238 NLL
- relative reduction: 52.28%

주의 :
This is intrinsic zero-step evidence, not after-MLM or downstream proof.

Source: docs/exp/v5/4_reporting/02_slides/final_deck_ko.md

Training setup

Main method claim 은 matched MLM checkpoint 이후에만 열린다 .

Fixed comparison:

- model: XLM-R-base initialized with v5 tokenizer
- methods: v5_random vs v5_fvt
- corpus/tokenizer/schedule/checkpoint rule 동일
- max steps: 10,000
- selected checkpoint: both 10K rows required

Current gate:

v5_random_mlm_10k: selected 10K checkpoint ready
v5_fvt_mlm_10k: running, model file pending
post-checkpoint evaluation: locked until wrapper-ready + preflight

Source:

docs/exp/v5/3_evaluation/running_status.md
docs/exp/v5/2_training/mlm_progress_eta.md

Speaker guard:

Live progress is not a model result; it only keeps the checkpoint gate honest.

Glot500 metrics

Glot500 에서 측정한 metric family 를 모두 유지한다 .
local coverage and target10 coverage:
• PPPL / MLM proxy: 102/102 local, target10 10/10
• Tatoeba retrieval: 63/102 local, target10 0/10
• Bible retrieval and Roundtrip alignment: 74/102 local, target10 0/10
• Text classification: 1/102 local, target10 0/10
• NER/POS: 78/102 audit and 58/102 local, target10 0/10
• v5 random PPPL/Tatoeba/Bible/Taxi1500/NER/POS/Roundtrip rows: measured;
FVT rows wait for matched checkpoints
Coverage principle:
missing task coverage is reported, not silently dropped.
Source:
docs/exp/v5/4_reporting/00_tables/table_13_metric_fidelity_matrix.md

Current measured rows

현재는 baseline/reference rows 와 v5-random PPPL/downstream rows 가 측정됐고 , FVT method claim 은 checkpoint 대기다 .

Measured baseline/reference rows:

- target PPPL: XLM-R 61.980216; Glot500-base 15.102934
- v5-random PPPL: target 39.222875; head 18.726452; all 20.138927
- Tatoeba all Top-10: XLM-R 0.566067; Glot500-base 0.706649; v5-random 0.610353
- Bible all Top-10: XLM-R 0.381153; Glot500-base 0.509356; v5-random 0.328019
- Taxi1500 macro-F1: XLM-R 0.592876; Glot500-base 0.743338; v5-random 0.702956
- NER all F1: XLM-R 0.549858; Glot500-base 0.627108; v5-random 0.544628
- POS all F1: XLM-R 0.481336; Glot500-base 0.567542; v5-random 0.481102
- Roundtrip acc.: XLM-R 0.185300; Glot500-base 0.205189; v5-random 0.190300

Interpretation:

v5-random is a diagnostic row, not a method win.
The decisive comparison is matched v5-random vs v5-FVT after the same 10K MLM budget.

Coverage and limitations

Target10 downstream coverage 는 현재 PPPL 중심이다 .

• PPPL target10 coverage: 10/10; retained downstream target10 coverage: 0/10

• NER target row is actual evaluated intersection, not target10-wide evidence

• Roundtrip 은 74/102 available-language 기준 XLM-R/Glot500-base/v5-random 측정 완료 , FVT row 는 checkpoint 대기

• dzo_Tibt tokenizer regression remains visible

Conclusion boundary:

target10 downstream improvement is not a supported claim yet.

Conclusion

현재 확정 결론은 setup fidelity + zero-step novelty 이다 .

Can say now:

• v5 reproduces the Glot500-style workflow on a controlled 102-language subset.
• tokenizer expansion and initialization audits are complete.
• FVT strongly improves zero-step target MLM proxy over random resize.
• all Glot500 metric families are retained with coverage/blocker accounting.

Cannot say yet:

• FVT improves after-MLM PPPL.
• FVT improves downstream performance.
• target10 downstream improves.

Final conclusion source:

docs/exp/v5/4_reporting/final_claim_decision_tree.md
docs/exp/v5/4_reporting/post_checkpoint_outcome_matrix_ko.md
docs/exp/v5/4_reporting/post_result_patch_plan_ko.md
docs/exp/v5/4_reporting/final_claim_freeze_audit.md

Backup: execution path

Checkpoint 이후 실행선

bash scripts/run_v5_post_checkpoint_evals.sh status
SKIP_MEASURED=1 WITH_PLOTS=1 GPU_RANDOM=0 GPU_FVT=1 bash scripts/run_v5_post_checkpoint_evals.sh all
python3 scripts/refresh_v5_reporting.py --with-plots
sed -n '1,120p' docs/exp/v5/4_reporting/post_result_patch_plan_ko.md

Audit files:

docs/exp/v5/4_reporting/final_handoff_runbook.md
docs/exp/v5/4_reporting/post_result_patch_plan_ko.md
docs/exp/v5/3_evaluation/post_checkpoint_execution_plan.md
docs/exp/v5/4_reporting/finalization_gate_status.md
docs/exp/v5/4_reporting/final_claim_freeze_audit.md
docs/exp/v5/4_reporting/reporting_package_audit.md
docs/exp/v5/4_reporting/artifact_reference_audit.md

Final rule:

Only aggregation-promoted values become final report/PPT claims.